

CCNP Lab: Multiple Wireless SSID with Access Point-WPA2 PSK, WPA2-Enterprise



By: Andy Yang

Purpose:

The purpose of this lab is to study and practice setting up additional access points for a certain Wifi/internet connection, so that said internet has a wider range, and can be accessed from further away from the main router. In this lab we set up a standalone Wireless Access Point with three different SSIDS, each with a different level of security. One with no security, which is open, one with a password, and a third one that requires authentication through a RADIUS server, which requires both a username and a password. We set up SSIDS each running on a different vlan, and through inter-vlan routing in a cisco switch to a router that is running router on a stick, which is connected to the internet.

Configuring multiple wireless SSIDS with Access-Point-WPA2 PSK and WPA2-Enterprise allows for distinct network configurations that can help suit different areas of use and user needs. It could also be especially useful in an environment where we want to segregate traffic for different purposes such as guest access, privileged mode, and admin access.

Background Information on lab concepts:

An Accesspoint is where we are creating our multiple distinct WIFI networks on. An Accesspoint is a networking device that acts as a bridge that allows wireless devices to wired networks. It also helps to extend wireless coverage and increase the number of users that can connect to a specific network.

SSIDS or Service Set Identifier which is a unique identifier for your WIFI network and is essence the name of your Wi-Fi-network and is used to help distinguish it from the other networks on your device.

VLANs, which stands for Virtual Local Area Network, is a local grouping of network devices that are treated as if they are in the same LAN, even if they are physically located on different parts of the network. In this lab specifically, the VLANs are used to create separate and isolated wireless networks, each with its own Service Set Identifier (SSID) and security configuration, all on the same physical network.

WPA2-PSK is a WIFI security protocol that uses a password to secure wireless networks, and encrypts data transmitted over the network, ensuring that only devices with the correct password can access it. In our lab specifically, it is used as a security protocol on one of our three Wi-Fi Networks.

WPA2-Enterprise is a WIFI security protocol that is an enhanced version of WPA2. It utilizes strong encryption and individualized user authentication that has distinct username and password for each individual user. In our lab specifically, it is used as a security protocol on one of our three Wi-Fi Networks.

DHCP or Dynamic Host Configuration Protocol is a protocol that helps to automatically assign IP addresses to devices connected to the network. DHCP in this lab specifically is used to help dynamically assign IP addresses to devices connecting to each of our three different VLANS associated with the SSIDS.

NAT or Network Address Translation – stands for a networking technique that allows multiple devices on a private network to share a single public IP address when connecting to the internet which is utilized in our lab as well to help conserve public IP addresses.

A Radius Server is a central server that provides remote authentication, authorization, and accounting (AAA) services for networking devices, connected through WIFI.

Lab Summary:

What was set up:

Router G0/0/1 connected to Cloud Internet.

Router G0/0/0 connected to Switch F0/19

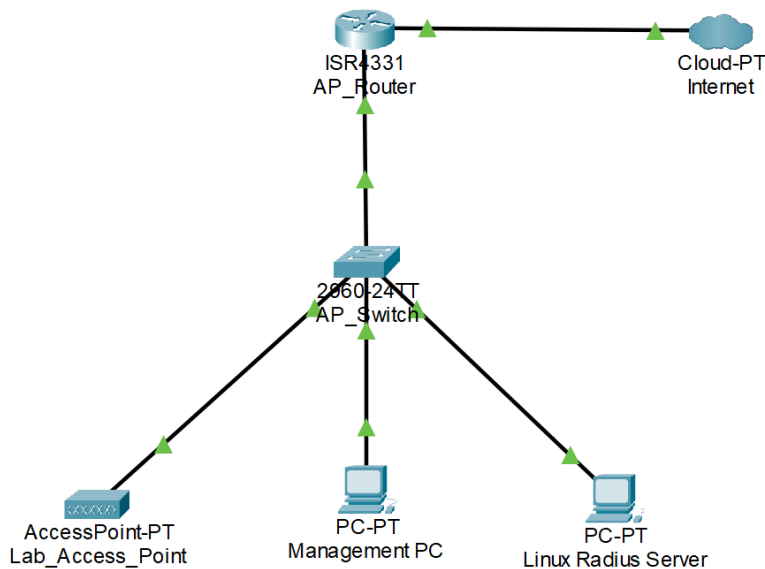
RG0/0/0/0.10, RG0/0/0.20, and RG0/0/0.30 sub interfaces are configured to for access to the Accesspoint, Management PC, and Linux Radius server.

Switch f0/17 connected to Accesspoint

Switch f0/19 connected to RADIUS server.

Switch f0/20 connected to PC.

Network Diagram:



Lab Commands:

show ip route – Displays the IPv4 routing table

ip routing – A command to enable the sending of data packets for IPv4 across an IP network

traceroute [destination ip address] - Figuring the data hops from the source ip address to the destination ip address.

Switchport mode access – Configures an interface to operate in access mode and carries traffic for only one VLAN.

Switchport trunk encapsulation dot1q – Configures an interface to operate in trunk mode and allows it to carry traffic for multiple vlans

Switchport trunk allowed vlan [vlan numbers separated by commas] - specifies which VLANs a trunk port is allowed to carry traffic for.

encapsulation dot1q [VLAN ID] - configures an interface to handle Ethernet frames with VLAN ID given.

ip dhcp pool [Pool_Name] - Creates a DHCP pool with the specified name which will define the parameters for IP address assignment on the network

Network [IP Address network] [Subnet Mask] - Specifies the IP address range that the DHCP pool will manage

Default-router [IP address] - Specifies the default gateway/router IP address that will be given to DHCP clients

Dns-server 8.8.8.8 - Gives the DNS server IP address that will be handed out to DHCP clients, with 8.8.8.8 is the public DNS server operated by Google.

ip dhcp excluded-address [ip address range] - Instructs a DHCP server to prevent assigning IP addresses from the ip range given to DHCP clients.

ip nat inside source list 1 interface [specified interface] overload – enables NAT on the specified Interface which helps to translate source addresses from the network defined in the access list

access-list 1 permit [IP address] [Wildcard-mask] - defines a network (192.168.1.0/24) for which NAT will be applied.

Configurations:

AP CLI Configuration:

```
version 15.3
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
hostname Lab-AP
```

```
logging rate-limit console 9
aaa new-model
aaa group server radius rad_eap
server name camera-ThinkStation-P7
aaa group server radius rad_mac
aaa group server radius rad_acct
server name camera-ThinkStation-P7
aaa group server radius rad_admin
server name camera-ThinkStation-P7
aaa group server tacacs+ tac_admin
aaa group server radius rad_pmip
aaa group server radius dummy
aaa authentication login eap_methods group rad_eap
aaa authentication login mac_methods local
aaa authorization exec default local
aaa accounting network acct_methods start-stop group rad_acct
aaa session-id common
no ip source-route
no ip cef
dot11 pause-time 100
dot11 syslog
dot11 ssid GuestNet
    vlan 20
    authentication open
    guest-mode
dot11 ssid Radius-CorpoNet
    vlan 30
    authentication open eap eap_methods
    authentication network-eap eap_methods
    authentication key-management wpa
    mbssid guest-mode
dot11 ssid SecureNet
    vlan 10
    authentication open
    authentication key-management wpa version 2
    mbssid guest-mode
    wpa-psk ascii 7 032752180500024340071C0603
no ipv6 cef
username Cisco password 7 0802455D0A16
bridge irb
interface Dot11Radio0
no ip address
ssid GuestNet
antenna gain 0
station-role root
bridge-group 1
bridge-group 1 subscriber-loop-control
bridge-group 1 spanning-disabled
bridge-group 1 block-unknown-source
no bridge-group 1 source-learning
no bridge-group 1 unicast-flooding
interface Dot11Radio0.20
```

```
encapsulation dot1Q 20
bridge-group 20
bridge-group 20 subscriber-loop-control
bridge-group 20 spanning-disabled
bridge-group 20 block-unknown-source
no bridge-group 20 source-learning
no bridge-group 20 unicast-flooding
interface Dot11Radio1
no ip address
encryption vlan 10 mode ciphers aes-ccm
encryption vlan 30 mode ciphers aes-ccm tkip
ssid Radius-CorpoNet
ssid SecureNet
antenna gain 0
peakdetect
dfs band 3 block
mbssid
channel dfs
station-role root
bridge-group 1
bridge-group 1 subscriber-loop-control
bridge-group 1 spanning-disabled
bridge-group 1 block-unknown-source
no bridge-group 1 source-learning
no bridge-group 1 unicast-flooding
interface Dot11Radio1.10
encapsulation dot1Q 10
bridge-group 10
bridge-group 10 subscriber-loop-control
bridge-group 10 spanning-disabled
bridge-group 10 block-unknown-source
no bridge-group 10 source-learning
no bridge-group 10 unicast-flooding
interface Dot11Radio1.30
encapsulation dot1Q 30
bridge-group 30
bridge-group 30 subscriber-loop-control
bridge-group 30 spanning-disabled
bridge-group 30 block-unknown-source
no bridge-group 30 source-learning
no bridge-group 30 unicast-flooding
interface Dot11Radio1.301
interface GigabitEthernet0
no ip address
duplex auto
speed auto
bridge-group 1
bridge-group 1 spanning-disabled
no bridge-group 1 source-learning
interface GigabitEthernet0.10
encapsulation dot1Q 10
bridge-group 10
```

```
bridge-group 10 spanning-disabled
no bridge-group 10 source-learning
interface GigabitEthernet0.20
encapsulation dot1Q 20
bridge-group 20
bridge-group 20 spanning-disabled
no bridge-group 20 source-learning
interface GigabitEthernet0.30
encapsulation dot1Q 30
bridge-group 30
bridge-group 30 spanning-disabled
no bridge-group 30 source-learning
interface GigabitEthernet0.301
interface BVI1
mac-address 44d3.ca03.7dce
ip address 192.168.1.245 255.255.255.0
ipv6 address autoconfig
ipv6 enable
ip default-gateway 192.168.0.1
ip forward-protocol nd
ip http server
no ip http secure-server
ip http help-path
http://www.cisco.com/warp/public/779/smbiz/prodconfig/help/eag
ip radius source-interface BVI1
radius-server local
radius-server attribute 32 include-in-access-req format %h
radius server camera-ThinkStation-P7
address ipv4 192.168.1.114 auth-port 1812 acct-port 1813
key 7 0538232C13697A
bridge 1 route ip
line con 0
line vty 0 4
transport input all
end
```

Cisco Switch CLI Configuration:

```
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
hostname APSwitch
boot-start-marker
boot-end-marker
no aaa new-model
system mtu routing 1500
authentication mac-move permit
ip subnet-zero
spanning-tree mode pvst
spanning-tree etherchannel guard misconfig
spanning-tree extend system-id
```

```
vlan internal allocation policy ascending
interface FastEthernet0/1
interface FastEthernet0/2
interface FastEthernet0/3
interface FastEthernet0/4
interface FastEthernet0/5
interface FastEthernet0/6
interface FastEthernet0/7
interface FastEthernet0/8
interface FastEthernet0/9
interface FastEthernet0/10
interface FastEthernet0/11
interface FastEthernet0/12
interface FastEthernet0/13
interface FastEthernet0/14
interface FastEthernet0/15
interface FastEthernet0/16
interface FastEthernet0/17
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 1,10,20,30
switchport mode trunk
interface FastEthernet0/18
switchport trunk encapsulation dot1q
switchport mode access
interface FastEthernet0/19
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 1,10,20,30
switchport mode trunk
interface FastEthernet0/20
switchport trunk encapsulation dot1q
switchport mode trunk
interface FastEthernet0/21
interface FastEthernet0/22
interface FastEthernet0/23
interface FastEthernet0/24
interface FastEthernet0/25
interface FastEthernet0/26
interface FastEthernet0/27
interface FastEthernet0/28
interface FastEthernet0/29
interface FastEthernet0/30
interface FastEthernet0/31
interface FastEthernet0/32
interface FastEthernet0/33
interface FastEthernet0/34
interface FastEthernet0/35
interface FastEthernet0/36
interface FastEthernet0/37
interface FastEthernet0/38
interface FastEthernet0/39
interface FastEthernet0/40
interface FastEthernet0/41
```



```
interface FastEthernet0/42
interface FastEthernet0/43
interface FastEthernet0/44
interface FastEthernet0/45
interface FastEthernet0/46
interface FastEthernet0/47
interface FastEthernet0/48
interface GigabitEthernet0/1
interface GigabitEthernet0/2
interface GigabitEthernet0/3
interface GigabitEthernet0/4
interface Vlan1
no ip address
ip classless
ip http server
ip http secure-server
ip sla enable reaction-alerts
line con 0
logging synchronous
line vty 0 4
login
line vty 5 15
login
end
```

Cisco Router CLI Configuration:

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname AP-Router
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp pool SECURE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

```
ip dhcp pool GUEST
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
ip dhcp pool RADIUS
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-2731081846
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2731081846
revocation-check none
rsakeypair TP-self-signed-2731081846
license udi pid ISR4321/K9 sn FD021432ZLS
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
negotiation auto
interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip nat inside
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip nat inside
interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
ip nat inside
interface GigabitEthernet0/0/1
ip address dhcp
ip nat outside
negotiation auto
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
```

```
no ip address
negotiation auto
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip http client source-interface GigabitEthernet0/0/0
ip nat inside source list 1 interface GigabitEthernet0/0/1 overload
access-list 1 permit 192.168.1.0 0.0.0.255
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
control-plane
line con 0
logging synchronous
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Radius Server on the AP Configuration:

Hostname: Lab-AP Lab-AP uptime is 1 day, 11 minutes

Security: Global SSID Manager

SSID Properties

Current SSID List: < NEW >
GuestNet
LabNet-Corporate
LabNet-Test

SSID:
 VLAN: Define VLANs

Band Select: ☐ Band Select
 Universal Admin Mode: ☐ Universal Admin Mode
☒ Radio-802.11n2-4GHz
☒ Radio-802.11n2-5GHz

Network ID: Delete

Client Authentication Settings

Methods Accepted:
☒ Open Authentication:
☐ Web-Auth:
☐ Shared Authentication:
☒ Network EAP:

Server Priorities:
EAP Authentication Servers
☒ Use Defaults Define Defaults
☐ Customise
 Priority 1:
 Priority 2:
 Priority 3:

MAC Authentication Servers
☒ Use Defaults Define Defaults
☐ Customise
 Priority 1:
 Priority 2:
 Priority 3:

Client Authenticated Key Management

Key Management: CCM ☒ Enable WPA

WPA Pre-shared Key:
 WPA Configuration:
 WPA Association-comeback: (1000-20000)
 WPA Sequence-entropy: (100-500)

Activate Windows
Go to Settings to activate Windows.

Hostname: Lab-AP Lab-AP uptime is 23 hours, 33 minutes

Security: Server Manager

Backup RADIUS Server

IP Version: ☒ IPv4 ☐ IPv6
 Backup RADIUS Server Name:
 Backup RADIUS Server: (Hostname or IP Address)
 Shared Secret:

Apply Delete Cancel

Corporate Servers

Current Server List: < NEW >
camera-ThinkStation-P7

IP Version: ☒ IPv4 ☐ IPv6
 Server Name:
 Server: (Hostname or IP Address)
 Shared Secret:

Authentication Port (optional): (0-65535)
 Accounting Port (optional): (0-65535)

Apply Cancel

Default Server Priorities

EAP Authentication
 Priority 1:
 Priority 2:
 Priority 3:

MAC Authentication
 Priority 1:
 Priority 2:
 Priority 3:

Accounting
 Priority 1:
 Priority 2:
 Priority 3:

Admin Authentication (RADIUS)
 Priority 1:
 Priority 2:
 Priority 3:

Admin Authentication (TACACS+)
 Priority 1:
 Priority 2:
 Priority 3:

Apply Cancel

Multiple BSSID Beacon Settings

Multiple BSSID Beacon

- ☒ Set SSID as Guest Mode
- ☐ Set DataBeacon Rate (DTIM): (1-100)

Guest Mode/Infrastructure SSID Settings

Radio0-802.11N2-4GHz:

- Set Beacon Mode: ☒ Single BSSID Set Single Guest Mode SSID:
☐ Multiple BSSID
- Set Infrastructure SSID: ☐ Force Infrastructure Devices to associate only to this SSID

Radio1-802.11N5-5GHz:

- Set Beacon Mode: ☐ Single BSSID Set Single Guest Mode SSID:
☒ Multiple BSSID
- Set Infrastructure SSID: ☐ Force Infrastructure Devices to associate only to this SSID

GuestNet configurations on the AP Configuration:

Hostname: Lab-AP

Security: Global SSID Manager

SSID Properties

Current SSID List

< NEW >

GuestNet
Radius-CorporateNet
SecureNet

SSID:

VLAN:

Band-Select:

Universal Admin Mode:

Interface:

GuestNet

20 [Define VLANs](#)

Backup 1:

Backup 2:

Backup 3:

☐ Band Select

☐ Universal Admin Mode

☒ Radio0-802.11N/40MHz

☐ Radio1-802.11N/GHz

Network ID: (0-4096)

[Delete](#)

Methods Accepted:

☒ Open Authentication:

☐ Web Authentication:

☐ Shared Authentication:

☐ Network EAP:

Server Priorities:

EAP Authentication Servers

☒ Use Defaults [Define Defaults](#)

☐ Customize

Priority 1:

Priority 2:

Priority 3:

MAC Authentication Servers

☒ Use Defaults [Define Defaults](#)

☐ Customize

Priority 1:

Priority 2:

Priority 3:

Key Management:

☐ CCKM ☐ Enable WPA

WPA Pre-shared Key:

☒ ASCII ☐ Hexadecimal

11w Configuration:

11w Association-comeback: (1000-20000)

11w Saquery-retry: (100-500)

Radio0-802.11N/40MHz:

Set Beacon Mode: ☒ Single BSSID Set Single Guest Mode SSID:

☐ Multiple BSSID

Set Infrastructure SSID: ☐ Force Infrastructure Devices to associate only to this SSID

Radio1-802.11N/GHz:

Set Beacon Mode: ☐ Single BSSID Set Single Guest Mode SSID:

☒ Multiple BSSID

Set Infrastructure SSID: ☐ Force Infrastructure Devices to associate only to this SSID

SecureNet configurations on the AP Configuration:

Hostname: Lab-AP

Lab-AP uptime is 1 day, 11 min

Security: Global SSID Manager

SSID Properties

Current SSID List

NEW +

GuestNet

Radius-Corporate

SecureNet

SSID:

SecureNet

VLAN:

10

Define VLANs

Backup 1:

Backup 2:

Backup 3:

Band Select:

☐ Band Select

Universal Admin Mode:

☐ Radio0-802.11N 40MHz

Interface:

☒ Radio1-802.11N 20MHz

Network ID:

(0-4096)

Delete

Client Authentication Settings

Methods Accepted:

☒ Open Authentication:

< NO ADDITION >

☐ Web Authentication:

< NO ADDITION >

☐ Shared Authentication:

< NO ADDITION >

☐ Network EAP:

< NO ADDITION >

Server Priorities:

EAP Authentication Servers

Use Defaults

Define Defaults

Customize

Priority 1:

< NONE >

Priority 2:

< NONE >

Priority 3:

< NONE >

MAC Authentication Servers

Use Defaults

Define Defaults

Customize

Priority 1:

< NONE >

Priority 2:

< NONE >

Priority 3:

< NONE >

Client Authenticated Key Management

Key Management:

Mandatory

WPA Pre-shared Key:

11w Configuration:

Disable

11w Association- comeback:

1000

(1000-20000)

11w Saquary-entry:

100

(100-500)

☐ COCK

☒ Enable WPA

WPA2

☒ ASCII

☐ Hexadecimal

Multiple BSSID Beacon Settings

Multiple BSSID Beacon

☒ Set SSID as Guest Mode

☐ Set DataBeacon Rate (DTIM): DISABLED

1-100

Guest Mode/Infrastructure SSID Settings

Radio0-802.11N 40MHz

Set Beacon Mode:

☒ Single BSSID

Set Single Guest Mode SSID: GuestNet

☐ Multiple BSSID

Set Infrastructure SSID:

< NONE >

☐ Force Infrastructure Devices to associate only to this SSID

Radio1-802.11N 20MHz

Set Beacon Mode:

☐ Single BSSID

Set Single Guest Mode SSID: < NONE >

☒ Multiple BSSID

Set Infrastructure SSID:

< NONE >

☐ Force Infrastructure Devices to associate only to this SSID

Apply

Cancel

Activate Windows

Go to Settings to activate Windows.

Radio0configurations on the AP Configuration:

AP Summary:

Hostname: Lab-AP					Lab-AP uptime is 1 day, 18 minutes
Network Interface: Summary					
System Settings					
IP Address (Static)		192.168.1.245			
IP Subnet Mask		255.255.255.0			
Default Gateway		192.168.1.1			
MAC Address		4403.c803.7000			
Interface Status					
Up/Ethernet		Radio-802.11b/g/n		Radio1-802.11b/g/n	
Software Status	Enabled		Enabled	Enabled	
Hardware Status	Up		Up	Up	
Interface Resets	1		2	19	
Receive					
Input Rate Timespan	5 minute		5 minute	5 minute	
Input Rate (bits/sec)	310000		0	24000	
Input Rate (packets/sec)	18		0	6	
Time Since Last Input	00:00:00		03:32:42	00:00:00	
Total Packets Input	570439		164277	244225	
Total Bytes Input	70911739		24372192	28440774	
Broadcast Packets	15761		1644	7831	
Total Input Errors	0		0	0	
Overrun Errors	0		0	0	
Ignored Packets	0		0	0	
Throttles	0		0	0	
Transmit					
Output Rate Timespan	5 minute		5 minute	5 minute	
Output Rate (bits/sec)	77000		0	310000	
Output Rate (packets/sec)	7		0	7	
Time Since Last Output	00:00:00		00:00:00	00:00:00	
Total Packets Output	276389		136513	382918	
Total Bytes Output	73126200		173644296	533651010	
Total Output Errors	0		10	22	
Last Output Hang	never		never	never	

SSIDS:

Home

Summary

Easy Setup

Network Configuration

Network Assistant

Hostname: Lab-AP

Lab-AP uptime is 1 day, 21 minutes

Reboot AP

Factory Reset

Network Configuration

Host Name:

Lab-AP

Server Protocol:

☐ DHCP
 ☒ Static IP

IP Address:

192.168.1.245

IP Subnet Mask:

255.255.255.0

Default Gateway:

192.168.1.1

IPv6 Protocol:

☐ DHCP
 ☒ Autoconfig
 ☐ Static IP

IPv6 Address:

(XXXX:XX:XX:XX:XX:XX)

Create a user

Username:

Password:

Change global authentication password

default enable secret:

confirm enable secret:

SNMP Community:

defaultCommunity
 ☒ Read-Only
 ☐ Read-Write

Apply

Cancel

Radio Configuration

Radio 2.4GHz

SSID:

GuestNet

☐ Broadcast SSID in Beacon
 ☒ Broadcast SSID in Beacon

VLAN:

☐ No VLAN
 ☒ Enable VLAN ID:

10
 ☐ (1-4094)
 ☐ Native VLAN

Universal Admin Mode:

(Disable)
 ☐ Native VLAN

Security:

No Security
 ☐ WPA2-PSK
 ☐ WPA2-Enterprise

Role in Radio Network:

Access Point

Optimize Radio Network:

Default

Antenna Extension:

Enable

Channel:

Least Congested

Power:

Maximum

Apply

Cancel

Radio 5GHz

SSID:

SecureNet

☐ Broadcast SSID in Beacon
 ☒ Broadcast SSID in Beacon

VLAN:

☐ No VLAN
 ☒ Enable VLAN ID:

10
 ☐ (1-4094)
 ☐ Native VLAN

Universal Admin Mode:

(Disable)
 ☐ Native VLAN

Security:

WPA2-PSK
 ☐ WPA2-Enterprise

Role in Radio Network:

Access Point

Optimize Radio Network:

Default

Antenna Extension:

Enable

Channel:

Dynamic Frequency Selection

Power:

Maximum

Apply

Cancel

Radio 2.4GHz

SSID:

GuestNet

☐ Broadcast SSID in Beacon
 ☒ Broadcast SSID in Beacon

VLAN:

☐ No VLAN
 ☒ Enable VLAN ID:

10
 ☐ (1-4094)
 ☐ Native VLAN

Universal Admin Mode:

(Disable)
 ☐ Native VLAN

Security:

No Security
 ☐ WPA2-PSK
 ☐ WPA2-Enterprise

Role in Radio Network:

Access Point

Optimize Radio Network:

Default

Antenna Extension:

Enable

Channel:

Least Congested

Power:

Maximum

Apply

Cancel

Radio 5GHz

SSID:

Double CapNet

☐ Broadcast SSID in Beacon
 ☒ Broadcast SSID in Beacon

VLAN:

☐ No VLAN
 ☒ Enable VLAN ID:

10
 ☐ (1-4094)
 ☐ Native VLAN

Universal Admin Mode:

(Disable)
 ☐ Native VLAN

Security:

WPA2-Enterprise
 ☐ WPA2-PSK

Role in Radio Network:

Access Point

Optimize Radio Network:

Default

Antenna Extension:

Enable

Channel:

Dynamic Frequency Selection

Power:

Maximum

Apply

Cancel

Radius Server commands:

```
cameron@cameron-ThinkStation-P7:~$ sudo nano /etc/freeradius/3.0/sites-enabled/default
cameron@cameron-ThinkStation-P7:~$ sudo systemctl restart freeradius
```

```
cameron@cameron-ThinkStation-P7:~$ sudo chmod -R 000 /etc/freeradius
[sudo] password for cameron:
cameron@cameron-ThinkStation-P7:~$ sudo systemctl restart freeradius
Job for freeradius.service failed because the control process exited with error
code.
See "systemctl status freeradius.service" and "journalctl -xeu freeradius.servic
e" for details.
cameron@cameron-ThinkStation-P7:~$ sudo systemctl enable freeradius\
>
Synchronizing state of freeradius.service with SysV service script with /usr/lib
/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable freeradius
cameron@cameron-ThinkStation-P7:~$ sudo systemctl enable freeradius
Synchronizing state of freeradius.service with SysV service script with /usr/lib
/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable freeradius
cameron@cameron-ThinkStation-P7:~$ █

cameron@cameron-ThinkStation-P7:~$ sudo chown -R freerad:freerad /etc/freeradius
sudo chown -R freerad:freerad /var/log/freeradius
sudo chmod -R 755 /etc/freeradius
[sudo] password for cameron:
cameron@cameron-ThinkStation-P7:~$ sudo system ctl restart freeradius
sudo: system: command not found
cameron@cameron-ThinkStation-P7:~$ sudo systemctl restart freeradius
cameron@cameron-ThinkStation-P7:~$ █
```

```
GNU nano 7.2
# -*- text -*-
##
## clients.conf -- client configuration directives
##
##      $Id: 5f39ff120a44e3fb837d1893d54a7ebb31fd8749 $
#####
#
# Define RADIUS clients (usually a NAS, Access Point, etc.).
client UserLink {
    ipaddr = 192.168.1.245
    secret = SECRET
}
#
# There are a number of security practices which are critical in the
# modern era.
#
# * don't use RADIUS/UDP or RADIUS/TCP over the Internet. Use RADIUS/TLS.
#
# * If you do send RADIUS over UDP or TCP, don't send MS-CHAPv2.
#   Anyone who can see the MS-CHAPv2 data can crack it in milliseconds.
#
# * use the "radsecret" program to generate secrets. It uses Perl (sorry).
```

Online Accounts

```
cameron@cameron-ThinkStation-P7: ~
GNU nano 7.2 /etc/freeradius/3.0/users *
testing Cleartext-Password := "password"

##
# Configuration file for the rlm_files module.
# Please see rlm_files(5) manpage for more information.
#
# This file contains authentication security and configur
# information for each user. Accounting requests are NOT
# through this file. Instead, see 'accounting', in this
#
# The first field is the user's name and can be up to
# 253 characters in length. This is followed (on the sam
# the list of authentication requirements for that user.
# include password, comm server name, comm server port nu
# time (perhaps set by the "hfile" file) and lastpasswa
```



```
cameron@cameron-ThinkStation-P7:~$
Mon Mar 31 14:35:23 2025 : Info: # Skipping contents of 'if' as it is always 'false' -- /etc/freeradius/3.0/sites-enabled/inner-tunnel:366
Mon Mar 31 14:35:23 2025 : Info: Loaded virtual server inner-tunnel
Mon Mar 31 14:35:23 2025 : Info: Ready to process requests
Mon Mar 31 14:35:58 2025 : Error: BlatRADIUS checks Received packet with Message-Authenticator.
Mon Mar 31 14:35:58 2025 : Error: Setting "require_message_authenticator = true" for client localhost
Mon Mar 31 14:35:58 2025 : Error: It looks like the client has been updated to protect from the BlatRADIUS attack.
Mon Mar 31 14:35:58 2025 : Error: Please set "require_message_authenticator = true" for client localhost
Mon Mar 31 14:35:58 2025 : Error: Signalled to terminate
Mon Mar 31 14:39:20 2025 : Info: Exiting normally
Mon Mar 31 14:39:20 2025 : Info: Debug state unknown (cap_sys_ptrace capability not set)
Mon Mar 31 14:39:20 2025 : Info: systemd watchdog interval is 30.00 secs
Mon Mar 31 14:39:20 2025 : Info: Loaded virtual server <default>
Mon Mar 31 14:39:20 2025 : Warning: Ignoring "sql" (see raddb/mods-available/README.rst)
Mon Mar 31 14:39:20 2025 : Warning: Ignoring "ldap" (see raddb/mods-available/README.rst)
Mon Mar 31 14:39:20 2025 : Info: Loaded virtual server default
Mon Mar 31 14:39:20 2025 : Info: # Skipping contents of 'if' as it is always 'false' -- /etc/freeradius/3.0/sites-enabled/inner-tunnel:366
Mon Mar 31 14:39:20 2025 : Info: Loaded virtual server inner-tunnel
Mon Mar 31 14:39:20 2025 : Info: Ready to process requests
Mon Mar 31 14:39:30 2025 : Error: BlatRADIUS checks Received packet with Message-Authenticator.
Mon Mar 31 14:39:30 2025 : Error: Setting "require_message_authenticator = true" for client localhost
Mon Mar 31 14:39:30 2025 : Error: It looks like the client has been updated to protect from the BlatRADIUS attack.
Mon Mar 31 14:39:30 2025 : Error: Please set "require_message_authenticator = true" for client localhost
Mon Mar 31 14:39:30 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:44 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:46 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:48 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:50 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:52 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:41:54 2025 : Error: Ignoring request to auth address * port 1812 bound to server default from unknown client 192.168.0.254 port 32770 proto udp
Mon Mar 31 14:44:27 2025 : Info: Signalled to terminate
Mon Mar 31 14:44:27 2025 : Info: Exiting normally
Mon Mar 31 14:44:28 2025 : Info: Debug state unknown (cap_sys_ptrace capability not set)
Mon Mar 31 14:44:28 2025 : Info: systemd watchdog interval is 30.00 secs
Mon Mar 31 14:44:28 2025 : Info: Loaded virtual server <default>
Mon Mar 31 14:44:28 2025 : Warning: Ignoring "sql" (see raddb/mods-available/README.rst)
Mon Mar 31 14:44:28 2025 : Warning: Ignoring "ldap" (see raddb/mods-available/README.rst)
Mon Mar 31 14:44:28 2025 : Info: Loaded virtual server default
Mon Mar 31 14:44:28 2025 : Info: # Skipping contents of 'if' as it is always 'false' -- /etc/freeradius/3.0/sites-enabled/inner-tunnel:366
Mon Mar 31 14:44:28 2025 : Info: Loaded virtual server inner-tunnel
Mon Mar 31 14:44:28 2025 : Info: Ready to process requests
cameron@cameron-ThinkStation-P7:~$ cd /
cameron@cameron-ThinkStation-P7:/$ cd etc
cameron@cameron-ThinkStation-P7:/etc$ sudo chmod -R 777 freeradius
[sudo] password for cameron:
cameron@cameron-ThinkStation-P7:/etc$ cd freeradius
cameron@cameron-ThinkStation-P7:/etc/freeradius$ ls
3.0
cameron@cameron-ThinkStation-P7:/etc/freeradius$ cd 3.0
cameron@cameron-ThinkStation-P7:/etc/freeradius/3.0$ ls
certs          huntgroups    policy.d      sites-enabled
clients.conf   mods-available proxy.conf    templates.conf
dictionary     mods-config   radiusd.conf  trigger.conf
experimental.conf mods-enabled  README.rst    users
hints          panic.gdb     sites-available
cameron@cameron-ThinkStation-P7:/etc/freeradius/3.0$
```

Show IP route/Traceroute for IPv4 and IPv6:

IPv4:

AP-Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user

static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -

LISP

a - application route

+ - replicated route, % - next hop override, p - overrides

from PFR

Gateway of last resort is 192.168.40.1 to network 0.0.0.0

S* 0.0.0.0/0 [254/0] via 192.168.40.1

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.1.1/32 is directly connected, GigabitEthernet0/0/0

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.10.0/24 is directly connected,

GigabitEthernet0/0/0.10

L 192.168.10.1/32 is directly connected,

GigabitEthernet0/0/0.10

192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.20.0/24 is directly connected,

GigabitEthernet0/0/0.20

L 192.168.20.1/32 is directly connected,

GigabitEthernet0/0/0.20

192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.30.0/24 is directly connected,

GigabitEthernet0/0/0.30

L 192.168.30.1/32 is directly connected,

GigabitEthernet0/0/0.30

C 192.168.40.0/23 is directly connected, GigabitEthernet0/0/1

192.168.40.0/32 is subnetted, 1 subnets

L 192.168.40.131 is directly connected, GigabitEthernet0/0/1

Traceroute/Pings:

```

C:\Users\Andy>ping 192.168.1.114

Pinging 192.168.1.114 with 32 bytes of data:
Reply from 192.168.1.114: bytes=32 time=1ms TTL=64
Reply from 192.168.1.114: bytes=32 time<1ms TTL=64
Reply from 192.168.1.114: bytes=32 time<1ms TTL=64
Reply from 192.168.1.114: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.114:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Andy>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Andy>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=2ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Andy>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Andy>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=2ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=2ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\Andy>

```

```

C:\Users\Andy>tracert 192.168.1.245

Tracing route to 192.168.1.245 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms    192.168.1.245

Trace complete.

C:\Users\Andy>

C:\Users\Andy>tracert 192.168.1.114

Tracing route to 192.168.1.114 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms    192.168.1.114

Trace complete.

```

```
C:\Users\Andy>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms    192.168.1.1

Trace complete.
```

PC pinging AP:

```
C:\Users\Andy>ping 192.168.1.245

Pinging 192.168.1.245 with 32 bytes of data:
Reply from 192.168.1.245: bytes=32 time<1ms TTL=255
Reply from 192.168.1.245: bytes=32 time<1ms TTL=255
Reply from 192.168.1.245: bytes=32 time<1ms TTL=255
Reply from 192.168.1.245: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.245:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Problems:

What was particularly challenging (half page):

There were various issues with simple layer 2 connectivity, where we were unable to ping directly connected routes, even if both sides had an ip address. This eventually led us to switch from a controller-based AP setup to a standalone AP, and because of the different GUIs that the controller and AP has, it was difficult to understand and learn what each part does. It was also the first time we ever tried to set up a Linux RADIUS server, and we were completely shooting in the dark, troubleshooting as we move forward.

Concepts we learned (half page):

We learned how to use an Accesspoint's GUI in order to create and manage multiple different SSIDs for network naming and identification. We also reviewed how to use DHCP on the router to help dynamically assign IP addresses to devices connecting to the different VLANs associated with the SSIDs. We also learned how to use NAT to set up in this lab in order to have multiple devices connect to the same IP address.

Another concept that we had to learn during this lab was the configuration of a Linux-based Radius Authentication Server.

What caused issues (half page):

We originally had a Controller as well as in our lab, but it ended up not being able to reach the Accesspoint and thus we ultimately changed to a standalone AP topology as we thought it would be simpler to configure. Another problem we ran into was trying to incorporate inter vlan-routing alongside router-on-a-stick configuration, which was something that we never tried before. We also had to learn the web-GUI of the Cisco WLAN Controller along with the web-GUI of the AP as we had to switch from a controller-based setup to a Standalone Access Point setup.

Conclusion:

Overall, this entire lab was a setup of multi-wireless APs in which there were 3 distinct Wi-Fi networks on an Access Point using VLANs, each with a unique authentication method and IP address. The first network is unencrypted and doesn't have a password, while the second uses WPA2-PSK which has a single shared password across all users. The third network uses WPA2-Enterprise, requiring individual usernames, passwords, and certificates, which is automatically authenticated by a RADIUS server. Lastly, a DHCP server automatically assigns IP addresses to devices connected to each VLAN's SSID.

Teacher Signoff Page of Lab Complete:

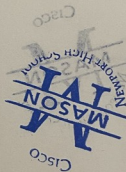
Configuring Wireless Access Point with WPA2 + PSK and Linux Radius Server Authentication Lab

Joshua Zhao P3-4 CCNP Mr. Mason



Configuring Wireless Access Point with WPA2 + PSK and Linux Radius Server Authentication Lab

Harrison Blizzard P3-4 CCNP Mr. Mason



Configuring Wireless Access Point with WPA2 + PSK and Linux Radius Server Authentication Lab

Andy Yang P3-4 CCNP

Mr. Mason

